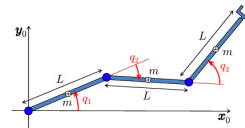


Exercise 1

Determine the expression of the control law $\tau = \tau(q, \dot{q})$ that realizes this behavior. For $L = 0.2$ [m], $m = 3$ [kg], $q_0 = (0, \pi/2, \pi/2)$ [rad], $\dot{q}_0 = (0, -\pi, -\pi)$ [rad/s] and $\gamma = 1$, compute the numerical value of such a control torque at $t = 0$, i.e., $\tau(0)$.

Consider the 3R robot in Fig. 1 moving on a horizontal plane. The robot has identical links (each of length L , uniformly distributed mass m , and inertia $I_L = mL^2/12$ around the barycentric vertical axis) and is commanded at the joint level by torques $\tau(t) \in \mathbb{R}^3$. Neglect in the following any dissipative/friction effects. With the system at $t = 0$ in a generic initial state $(q(0), \dot{q}(0)) = (q_0, \dot{q}_0)$ with $\dot{q}_0 \neq 0$, we want to control the robot so that its kinetic energy $T = T(q, \dot{q})$ in the closed-loop dynamics satisfies the following desired target equation:

$$\frac{dT}{dt} = -\gamma T, \quad \text{with } \gamma > 0.$$



We want the kinetic energy to exponentially go to zero.

$$p_{c1} = \begin{cases} \frac{1}{2} c_1 \\ \frac{1}{2} s_1 \end{cases} \quad u_{c1} = \begin{cases} -\frac{1}{2} s_1 \dot{q}_1 \\ \frac{1}{2} c_1 \dot{q}_1 \end{cases} \quad T_1 = \frac{1}{2} I \dot{q}_1^2 + \frac{1}{2} m \frac{L^2}{4} \dot{q}_1^2$$

$$p_{c2} = \begin{cases} L c_1 + \frac{1}{2} c_{12} \\ L s_1 + \frac{1}{2} s_{12} \end{cases} \quad u_{c2} = \begin{cases} -L s_1 \dot{q}_1 - \frac{1}{2} s_{12} (\dot{q}_1 + \dot{q}_2) \\ L c_1 \dot{q}_1 + \frac{1}{2} c_{12} (\dot{q}_1 + \dot{q}_2) \end{cases} \quad T_2 = \frac{1}{2} I (\dot{q}_1 + \dot{q}_2)^2 + \frac{1}{2} m \left(L^2 \dot{q}_1^2 + \frac{L^2}{4} (\dot{q}_1 + \dot{q}_2)^2 + 2 \frac{L^2}{2} c_2 (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2) \right)$$

$$u_{c3} = \begin{cases} -L s_1 \dot{q}_1 - L s_{12} (\dot{q}_1 + \dot{q}_2) - \frac{1}{2} s_{23} (\dot{q}_1 + \dot{q}_2 + \dot{q}_3) \\ L c_1 \dot{q}_1 + L c_{12} (\dot{q}_1 + \dot{q}_2) + \frac{1}{2} c_{23} (\dot{q}_1 + \dot{q}_2 + \dot{q}_3) \end{cases}$$

$$T_3 = \frac{1}{2} I (\dot{q}_1 + \dot{q}_2 + \dot{q}_3)^2 + \frac{1}{2} m \left[L^2 \dot{q}_1^2 + L^2 (\dot{q}_1 + \dot{q}_2)^2 + \frac{L^2}{4} (\dot{q}_1 + \dot{q}_2 + \dot{q}_3)^2 + 2 L^2 c_2 (\dot{q}_1^2 + \dot{q}_1 \dot{q}_2) + 2 \frac{L^2}{2} c_{23} (\dot{q}_1^2 + \dot{q}_2 \dot{q}_1 + \dot{q}_1 \dot{q}_3) + 2 \frac{L^2}{2} c_3 (\dot{q}_1^2 + 2 \dot{q}_1 \dot{q}_2 + \dot{q}_1 \dot{q}_3 + \dot{q}_2^2 + \dot{q}_2 \dot{q}_3) \right]$$

$$M_{11} = I + \frac{mL^2}{4} + I + mL^2 + m \frac{L^2}{4} + L^2 c_2 + I + mL^2 + mL^2 + m \frac{L^2}{4} + 2L^2 c_2 + L^2 c_{23} + L^2 c_3$$

$$= 3I + 3.75mL^2 + m \frac{1}{4} L^2 c_2 + mL^2 c_{23} + mL^2 c_3 = 3e_1 + 3.75e_2 + e_2 (4c_2 + c_3 + c_{23})$$

$$e_1 = I \quad e_2 = mL^2$$

$$M_{12} = I + \frac{mL^2}{4} + m \frac{L^2}{2} c_2 + I + mL^2 + m \frac{L^2}{4} + mL^2 c_2 + m \frac{L^2}{2} c_{23} + mL^2 c_3 =$$

$$2e_1 + 1.5e_2 + e_2 \left(\frac{3}{2} c_2 + \frac{1}{2} c_{23} + c_3 \right)$$

$$M_{13} = I + \frac{mL^2}{4} + mL^2 \frac{1}{2} c_{23} + mL^2 \frac{1}{2} c_3 = e_1 + \frac{1}{2} e_2 (c_3 + c_{23}) + \frac{1}{4} e_2$$

$$M_{22} = 2I + m \frac{L^2}{4} + m \frac{L^2}{2} c_2 + mL^2 + m \frac{L^2}{4} + mL^2 c_3 = 2e_1 + 1.5e_2 + e_2 \left(\frac{1}{2} c_2 + c_3 \right)$$

$$M_{23} = I + m \frac{L^2}{4} + mL^2 \frac{1}{2} c_3 = e_1 + \frac{1}{4} e_2 + \frac{1}{2} e_2 c_3$$

$$M_{33} = I + m \frac{L^2}{4} = e_1 + \frac{1}{4} e_2$$

The dynamic model in absence of gravity is

$$M\ddot{q} + c = \tau$$

We know that $T = \frac{1}{2} \dot{q}^T M \dot{q}$ so:

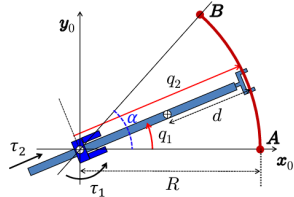
$$\frac{dT}{dt} = \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T M \ddot{q} \quad \text{and we want:}$$

$$\frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T M \ddot{q} = -\delta \frac{1}{2} \dot{q}^T M \dot{q} \quad \text{so we apply FBL}$$

$$M \ddot{q} + C = \tau, \quad \tau = C + a \Rightarrow M \ddot{q} = a \quad \text{then since we}$$

want

$$\dot{q}^T M \ddot{q} + \dot{q}^T \dot{M} \dot{q} = \frac{1}{2} \dot{q}^T (\delta M + \dot{M}) \dot{q} \Rightarrow$$

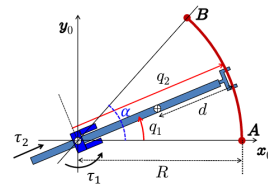


$$\dot{q} \quad \text{so we use:}$$

$$\tau = C + \frac{1}{2} (\delta M + M) \dot{q}$$

Exercise 2

The RP planar robot shown in Fig. 2 should execute a rest-to-rest motion task in minimum time under torque/force bounds $|\tau_i| \leq \tau_{max,i} > 0, i = 1, 2$, with its end-effector moving along a circular path of radius $R > d$ by an angle α from A to B. Determine the analytic expression of the minimum time T^* in terms of the task data and of the robot dynamic parameters. Draw the profile of the two components of the time-optimal command $\tau^*(t)$, for $t \in [0, T^*]$.



It is not specified, so we assume there is no gravity.

$$p_{c2} = \begin{cases} (q_2 - d) c_1 \\ (q_2 - d) s_1 \end{cases} \quad \dot{p}_{c2} = \begin{cases} -(q_2 - d) s_1 \dot{q}_1 + \dot{q}_2 c_1 \\ (q_2 - d) c_1 \dot{q}_1 + \dot{q}_2 s_1 \end{cases} \Rightarrow T = \frac{1}{2} (I_1 + I_2) \dot{q}_1^2 + \frac{1}{2} m_2 [(q_2 - d)^2 \dot{q}_1^2 + \dot{q}_2^2]$$

$$M = \begin{pmatrix} I_1 + I_2 + m_2 (q_2 - d)^2 & 0 \\ 0 & m_2 \end{pmatrix} \quad g = g_0 m_2 (q_2 - d) s_1 \Rightarrow g = \begin{cases} g_0 m_2 (q_2 - d) c_1 \\ g_0 m_2 s_1 \end{cases}$$

$$C_1 = \begin{pmatrix} 0 & m_2 (q_2 - d) \\ m_2 (q_2 - d) & 0 \end{pmatrix} \Rightarrow c_1 = 2 m_2 (q_2 - d) \dot{q}_1 \dot{q}_2$$

$$C_2 = -\frac{1}{2} \begin{pmatrix} 2 m_2 (q_2 - d) & 0 \\ 0 & 0 \end{pmatrix} = m_2 (d - q_2) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow C_2 = -m_2 (q_2 - d) \dot{q}_1^2$$

$$M \ddot{q} + C = \tau \Rightarrow \ddot{q} = \dot{M}^{-1} (\tau - C)$$

$$\ddot{q}_{max} = \frac{1}{(I_1 + I_2 + m_2 (q_2 - d)^2)} \begin{pmatrix} m_2 & 0 \\ 0 & I_1 + I_2 + m_2 (q_2 - d)^2 \end{pmatrix} \begin{pmatrix} \tau_{max,1} - 2 m_2 (q_2 - d) \dot{q}_1 \dot{q}_2 \\ \tau_{max,2} + m_2 (q_2 - d) \dot{q}_1^2 \end{pmatrix}$$

$$\ddot{q}_{max,2} = \tau_{max,2} + m_2(q_2 - d)\dot{q}_1^2 \quad \text{we have to set } q_2 = R$$

$$\Rightarrow \tau_{max,2} + m_2(R - d)\dot{q}_1^2 \quad \text{this should be kept zero}$$

$$\Rightarrow \tau_{max,2} = -m_2(R - d)\dot{q}_1^2 \Rightarrow \dot{q}_1 \leq \sqrt{\frac{\tau_{max,2}}{m_2(R - d)}} \quad \dot{q}_1 \text{ is limited by}$$

$$\text{this.} \quad \ddot{q}_{max,1} = \frac{\tau_{max,1} - 2m_2(q_2 - d)\dot{q}_1\dot{q}_2}{(I_1 + I_2 + m_2(q_2 - d)^2)} m_2$$

Exercise 3

With reference to Fig. 3 a mass m_1 is moving at constant speed $v_0 > 0$ and collides at some time $t = t_c$ with a mass m_2 which is initially at rest. Assume a purely ideal situation: there is no dissipation due to friction and the collision is perfectly elastic. Therefore, the total kinetic energy T and the total (scalar) momentum P of the two masses will both remain constant over time. Determine the expressions of the velocities $v_1(t_c^+)$ and $v_2(t_c^+)$ of the two masses after the collision. Describe what happens when $m_1 > m_2$, $m_1 = m_2$, or $m_1 < m_2$, and in the limit cases when $m_2 \rightarrow 0$ or $m_2 \rightarrow \infty$.

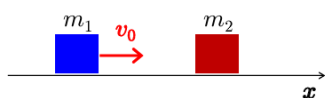


Figure 3: A mass m_1 in motion collides with a second mass m_2 initially at rest.

The initial kinetic energy is $\frac{1}{2}m_1 v_0^2$, we rewrite it as the sum of the energies of the two bodies: $T = \frac{1}{2}m_1 v_{01}^2 + \frac{1}{2}m_2 v_{02}^2$ where $v_{02}^2 = 0$ the two scalar momentum are $m_1 v_{01} + m_2 v_{02}$. After the impact we have the second body moving at v_{12}

$$\begin{cases} \frac{1}{2}m_1 v_{01}^2 + \cancel{\frac{1}{2}m_2 v_{02}^2} = \frac{1}{2}m_1 v_{11}^2 + \frac{1}{2}m_2 v_{12}^2 \\ m_1 v_{01} + \cancel{m_2 v_{02}} = m_1 v_{11} + m_2 v_{12} \end{cases} \quad \text{conservation}$$

I solve for v_{11} and v_{12} (velocities after hit)

$$\Rightarrow v_{12} = \frac{1}{m_2} (m_1 v_{01} - m_1 v_{11})$$

$$\frac{1}{2}m_1 v_{11}^2 = \frac{1}{2}m_1 v_{01}^2 - \frac{1}{2} \frac{1}{m_2} (m_1 v_{01} - m_1 v_{11})^2$$

$$v_{11}^2 = v_{01}^2 - \frac{1}{m_1 m_2} (m_1^2 v_{01}^2 + m_1^2 v_{11}^2 - 2m_1^2 v_{01} v_{11})$$

$$\left(1 + \frac{m_1}{m_2}\right) v_{11}^2 + 2 \frac{m_1}{m_2} v_{01} v_{11} + v_{01}^2 \left(\frac{m_1}{m_2} - 1\right) = 0 \quad \text{i solve}$$

for v_{11}

$$\Delta = 4 \frac{m_1^2}{m_2^2} v_{01}^2 - 4 v_{01}^2 \left(\frac{m_1^2}{m_2^2} - 1\right) = 4 v_{01}^2$$

$$v_{11} = \frac{-2 \frac{m_1}{m_2} v_{01} \pm 2 v_{01}}{2 + 2 \frac{m_1}{m_2}} = \frac{\pm 2 v_{01} \left(1 - \frac{m_1}{m_2}\right)}{2 + 2 \frac{m_1}{m_2}} \quad \text{let } \frac{m_1}{m_2} = \gamma$$

$$v_{11} = \frac{\pm 2 v_{01} (1 - \gamma)}{2(\gamma + 1)} \quad v_{12} = \gamma (v_{01} - v_{11})$$

if $m_1 = m_2$, $\gamma = 1$ hence $v_{11} = 0$ $v_{12} = v_{01}$ (the first stop, the second moves with the first body initial energy)

if $m_1 > m_2$ $\gamma > 1$ hence $v_{11} = \frac{\pm v_{01}(1-\gamma)}{2(\gamma+1)}$ we take the pos

v_{11} will be less than v_{01} , and the second body will move with a constant speed $v_{12} = \gamma(v_{01} - v_{11})$ greater than the original v_{01}

If $m_2 > m_1$, $\gamma < 1$ hence v_{11} will be negative and v_{12} positive.

if $m_2 \rightarrow \infty$, $\gamma \rightarrow 0$ hence